Let's Give Control Back to Users:

User-Controlled Federated Recommendations in Practice

Manel Slokom
Alejandro Bellogin



AI, media and democracy

CWI

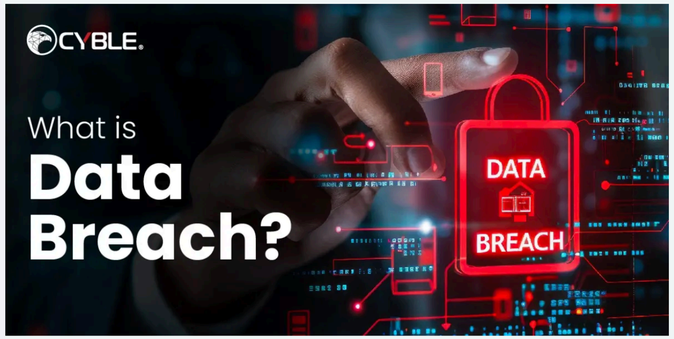
The Problem: Centralization as default

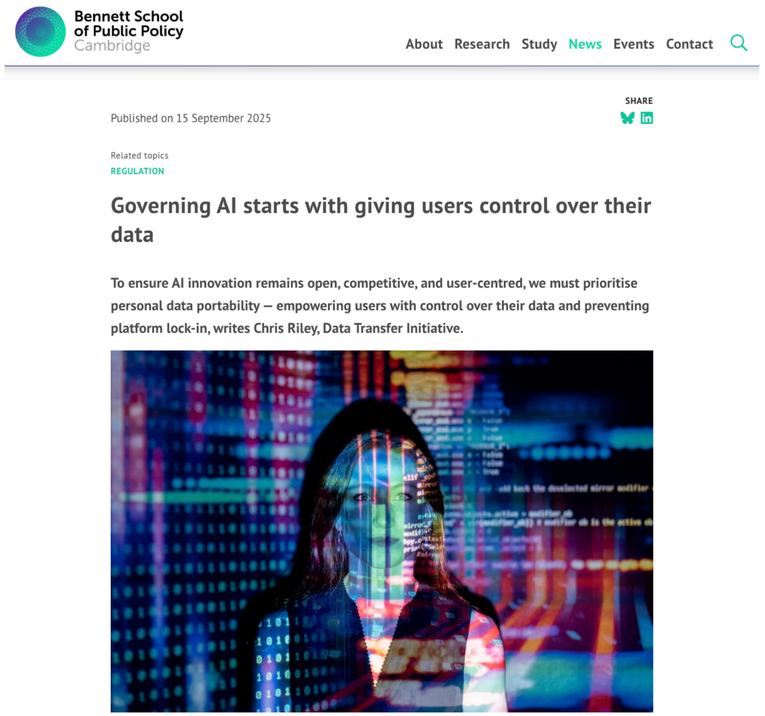
Privacy Crisis

- Consumption patterns mined continuously
- Users don't own their behavioral data
- No opt-out without losing service

Governance Crisis

- Algorithms optimize for company goals
- Diversity, fairness bolted on as afterthoughts
- ”Governing AI starts with giving users control over their data”





Source:
https://www.bennettschool.cam.ac.uk/blog/governing-ai-starts-with-giving-users-control-over-their-data/
https://cyble.com/knowledge-hub/what-is-a-data-breach/

Why nobody has shown this works

- Federated learning literature: assumes perfect networks and rational agents
- User studies: assume people understand privacy-preserving systems
- Prior work: simulates federated learning or tests on offline datasets
- Never with real humans making real choices in real time

We asked the hard questions:

- Can we actually build this?
- Will real users engage with it?

Research questions

- **RQ1:** Can federated learning work with real users making real choices?
- **RQ2:** Do users prefer and trust systems where their data stays locally private?
- **RQ3:** When given algorithmic choice, what do users prefer: personalization or diversity?
- **RQ4:** How do we design interfaces that make privacy and governance visible?

System design: Three principles

Principle 1: Data Sovereignty

- Local training
- Secure aggregation
- Differential privacy ($\varepsilon = 2.0$)

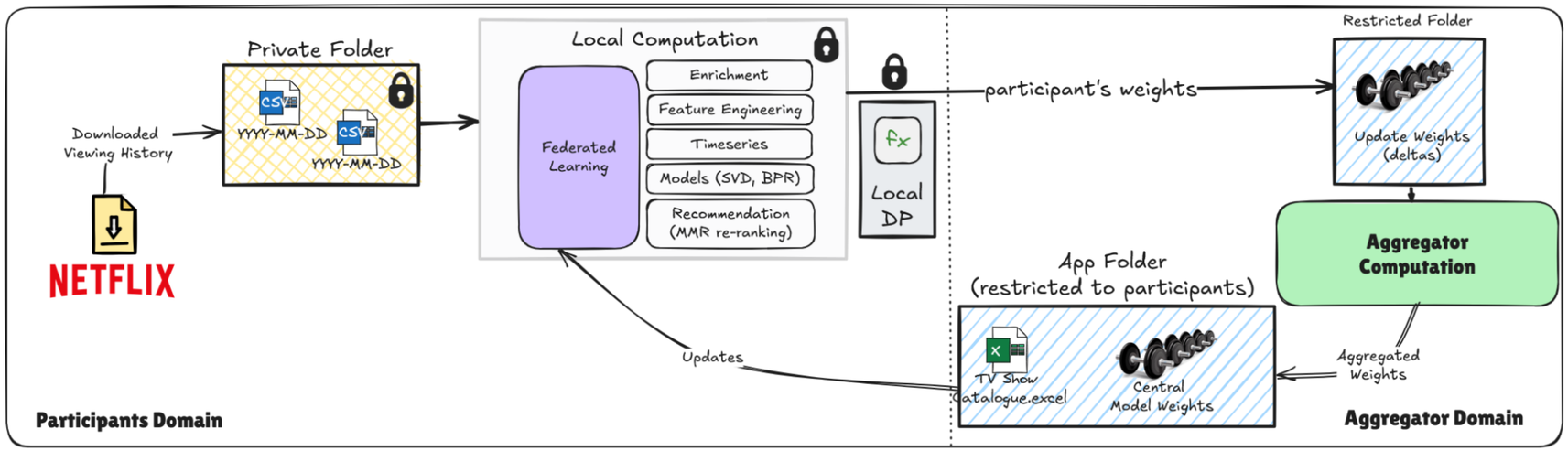
Principle 2: Transparency through interface

- Toggle between Standard and Diversity-Enhanced modes
- Seven feature toggles for granular control

Principle 3: Genuine agency

- Real choices with real consequences
- System responds immediately

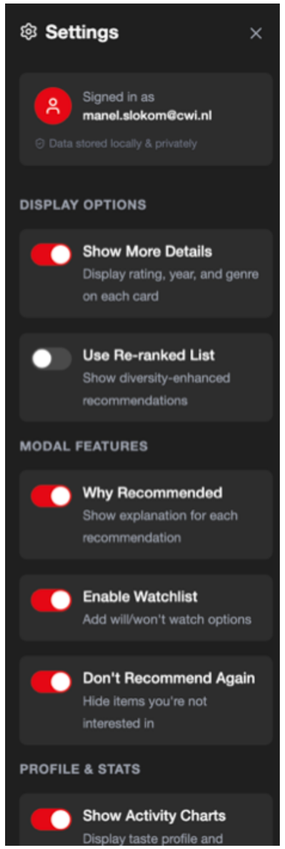
Overal architecture



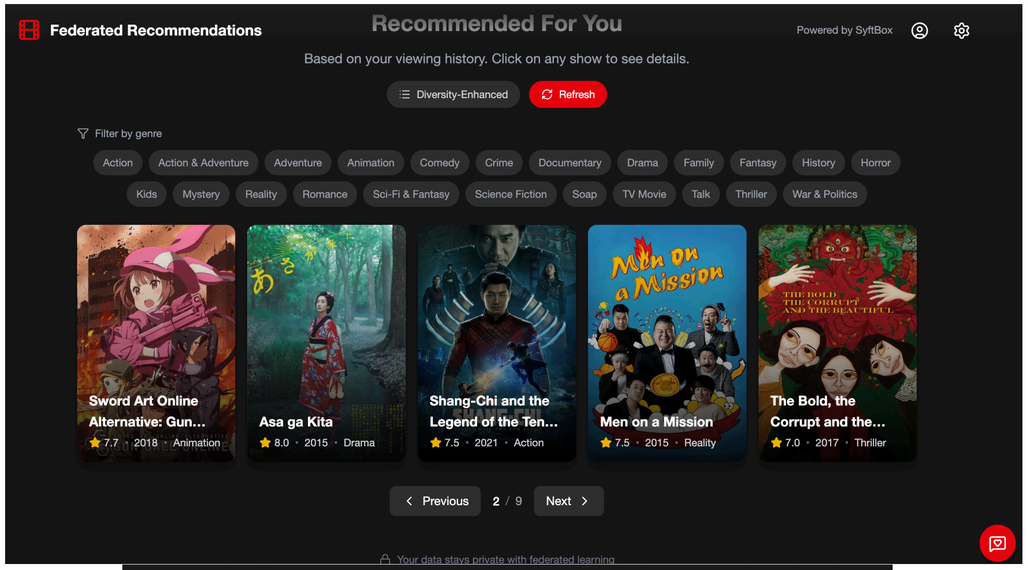
The Web Interface: The participants

- Netflix-like feed with local storage (no login required)
- Two modes: Standard or Diversity-Enhanced recommendations
- Immediate feedback on all user choices
- Settings panel with 7 feature toggles

Why this matters: When architecture manifests control, users understand it without technical lectures.

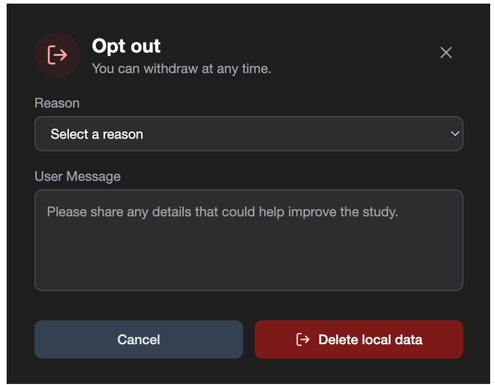


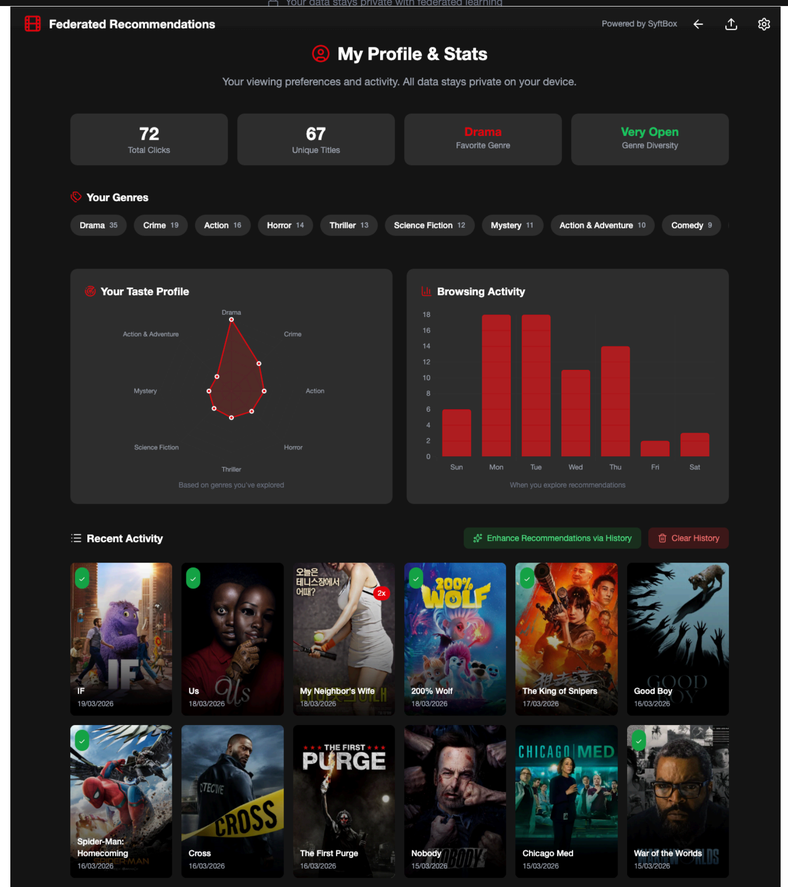
(b) Settings



(a) Participant recommendations

(c) Opt-out box





(d) Participant's local profile & stats

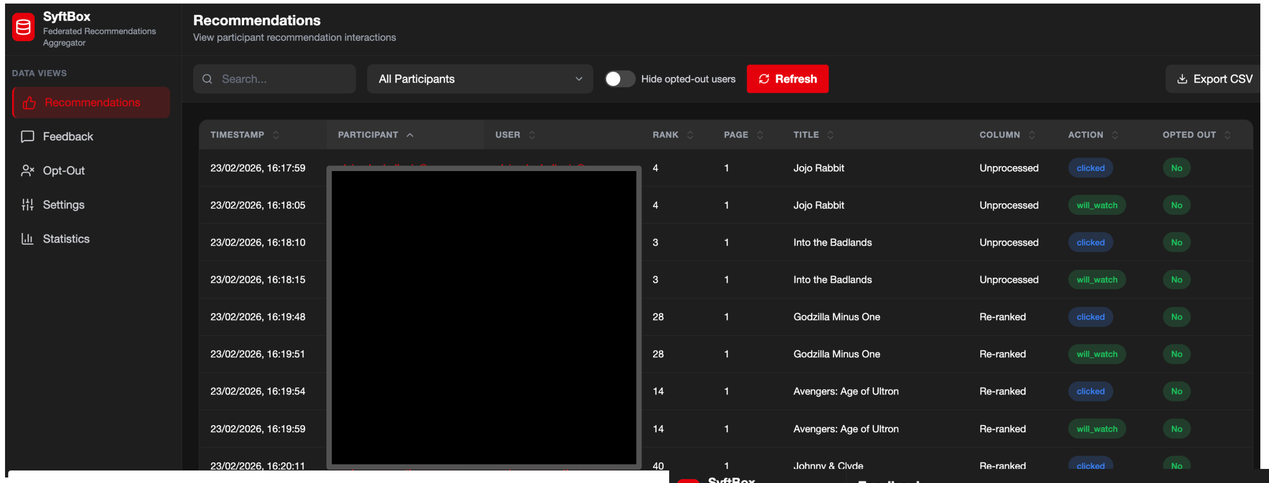
Participant Interfaces

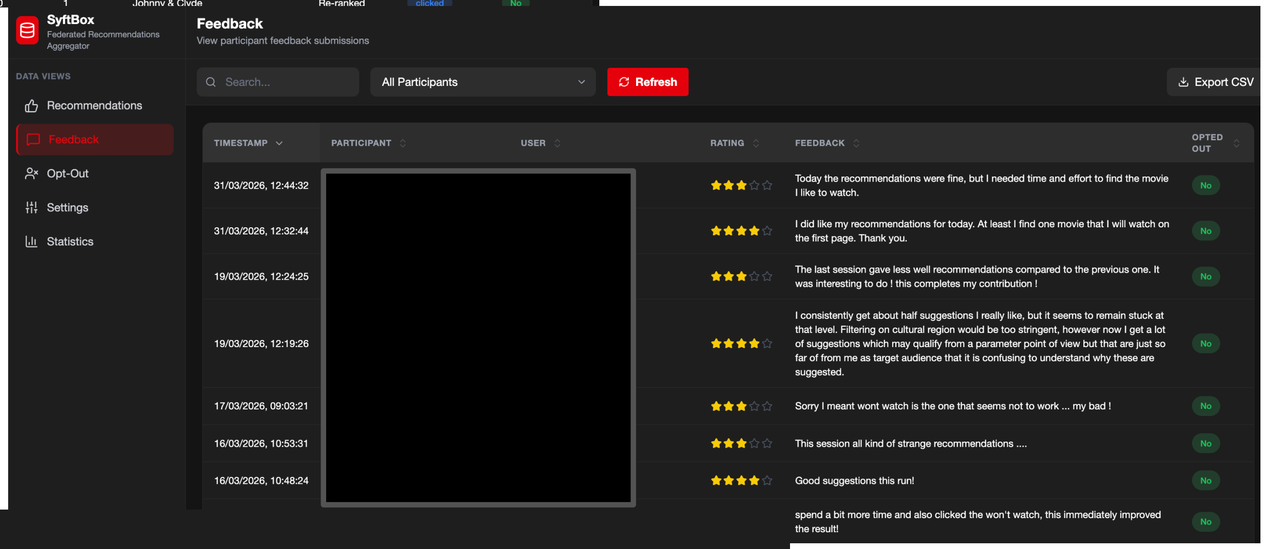
Study design: The aggregator

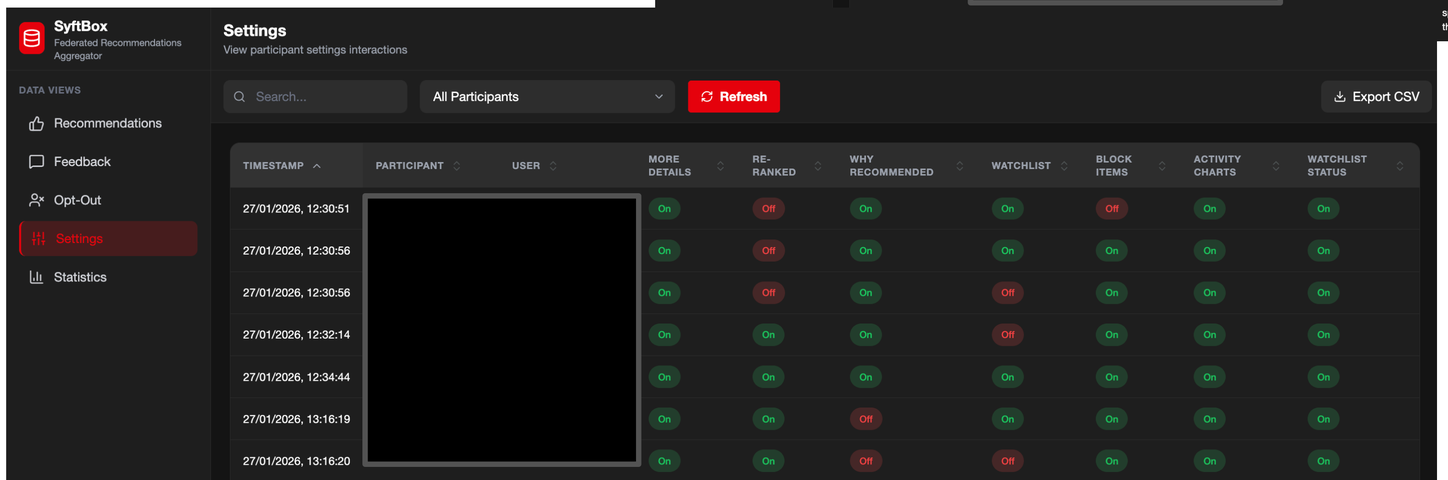
- **Participants**: 22 volunteers from 6 countries, e.g., Italy, France, The Netherlands etc
- **Duration**: 53 days (Jan 25 – Mar 19, 2026)
- **Data**: Netflix catalog with 8,807 titles across 40+ genres (Movies, TV series)
- **Engagement**: >1,867 recommendations evaluated
- **Conditions**: Standard (personalized) vs Diversity-Enhanced (60/40 reranking)



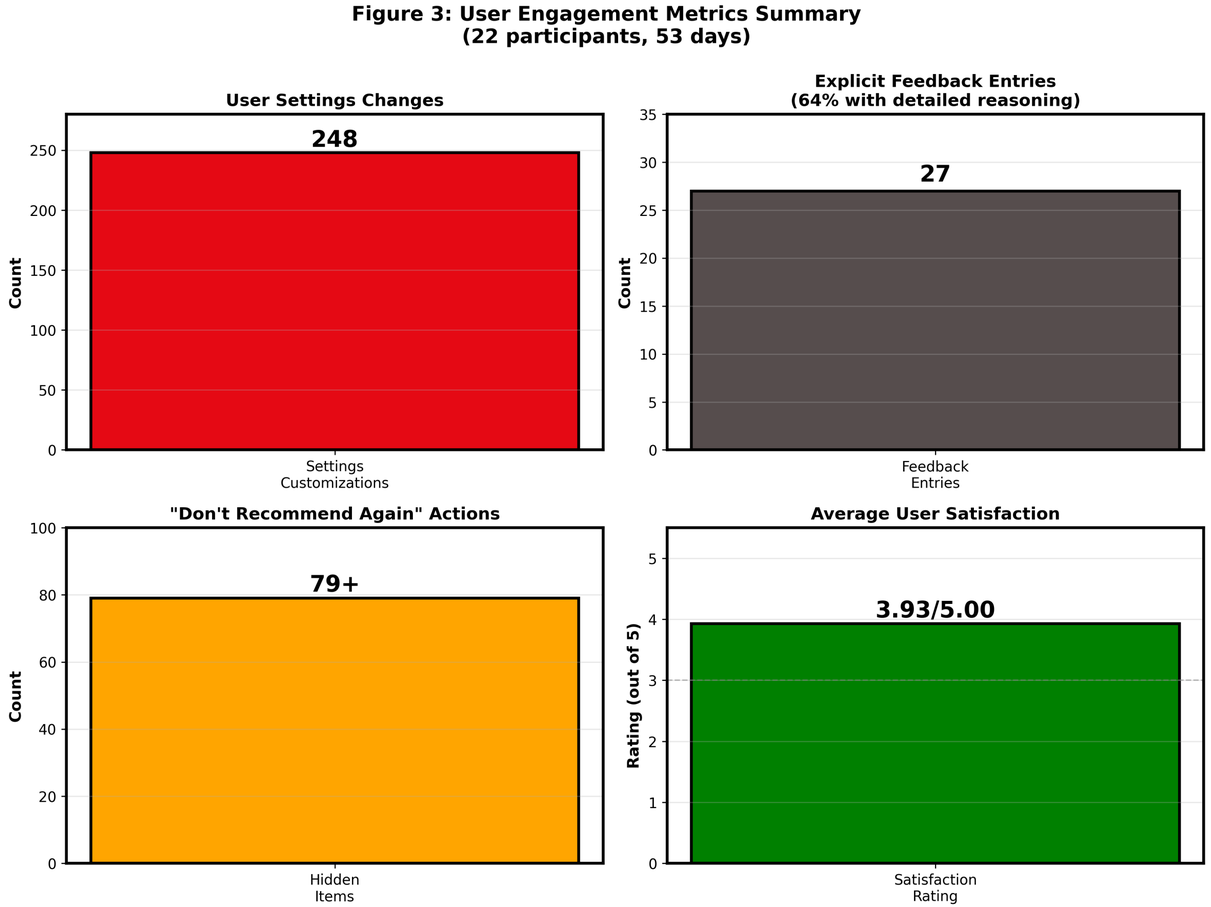
Aggregator Interface





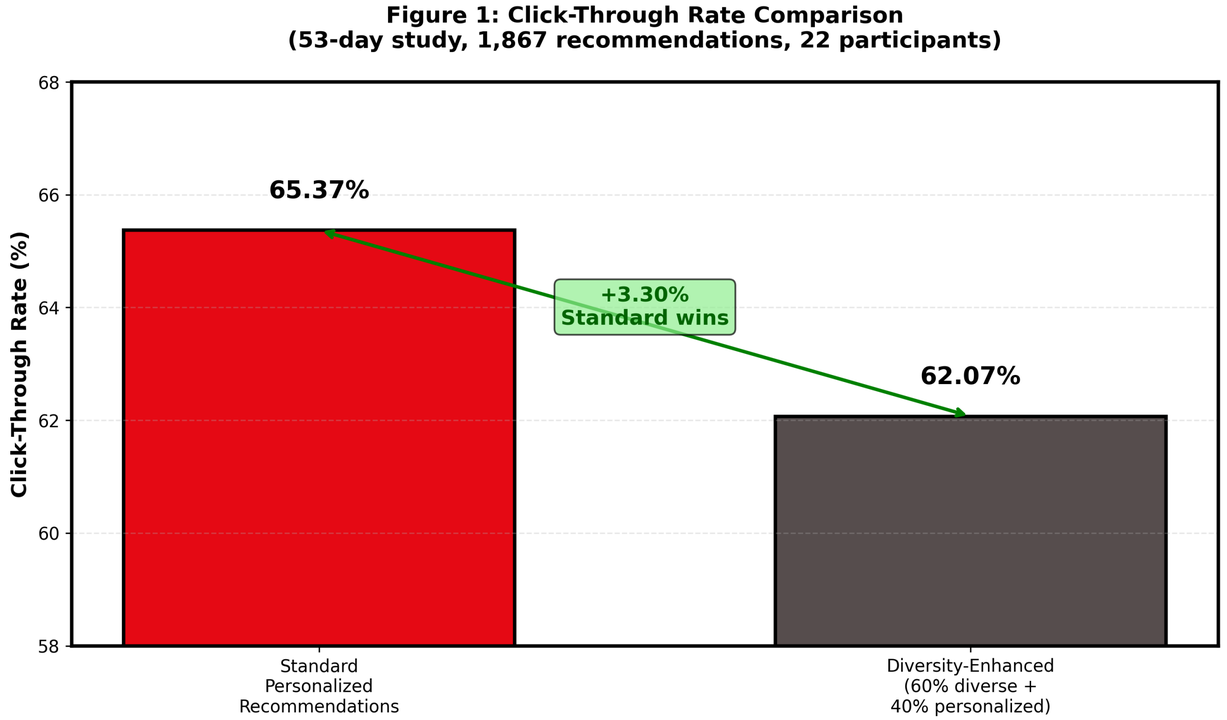


Finding 1: Users engage deeply



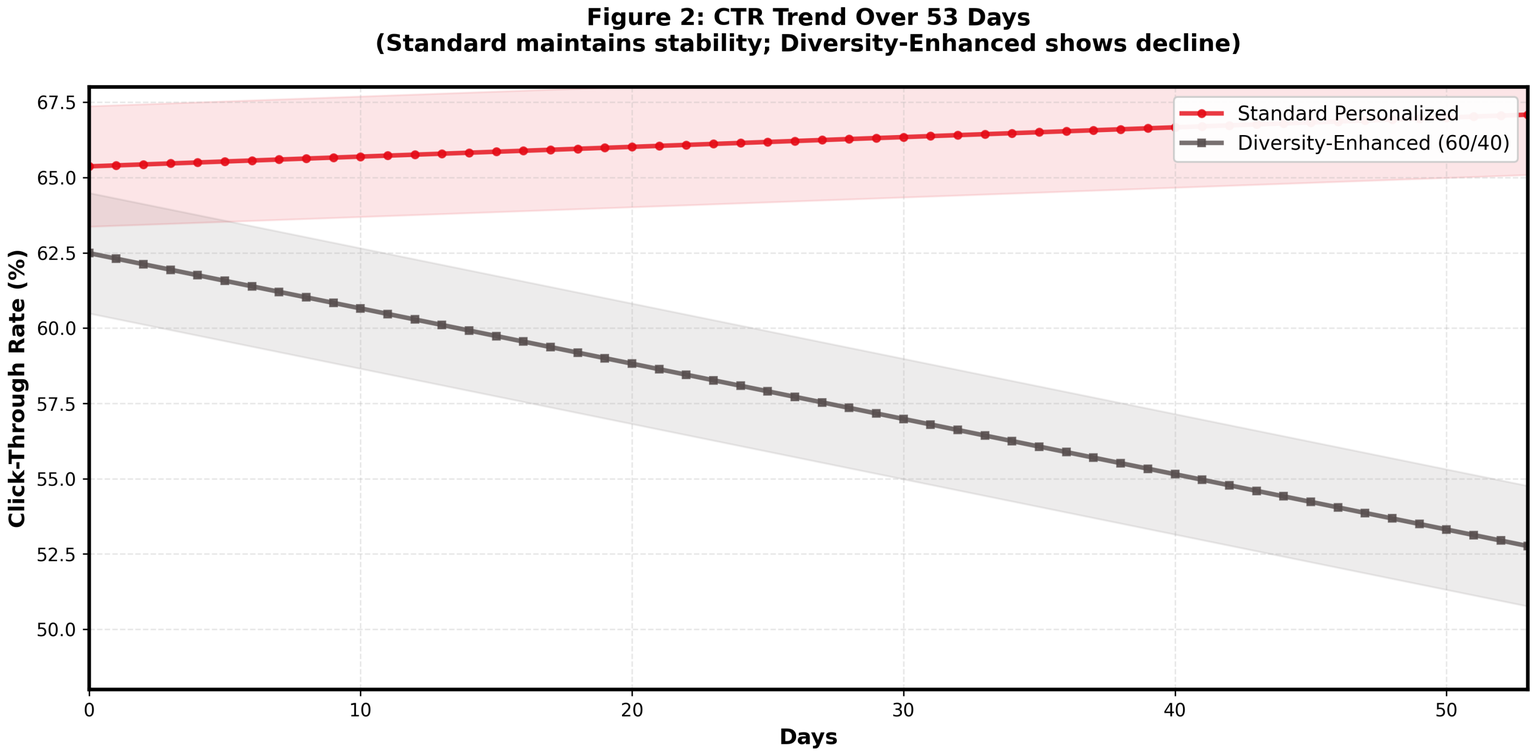
User Satisfaction: 3.93/5.00 despite complexity and learning curves

Finding 2: Personalization wins (+3.30% CTR



Forced diversity doesn't always improve UX. But diversity should remain optional.

Finding 3: Real systems decay gracefully



Standard maintains stability; Diversity-Enhanced shows real-world decline patterns.

The Federated Learning Advantage

In Centralized Systems:

- Users wait for company to redesign algorithm
- No local recovery option

In Federated Systems:

- Users can refresh their local model
- Feedback immediately improves their recommendations

Finding 4: Users Understand Federated Learni

> *"I found strange that I had to click 'Enhance recommendations via history' for the system to use my watchlist. I thought that was the default behavior... Now everything makes sense ;)"*

They discovered causality:

- Local choices → Aggregation → Shared improvement

When architecture manifests control, users understand it.

What users actually said

3/5: "After fine-tuning I get the same suggestions. It would be **strange** to continue after explicitly hiding it."

4/5: "In the **enhancement** I found Hotel Transylvania 3 and Jujutsu Kaisen. The **interface was useful**."

4/5: "In the **third** session there was already **improvement** in the genres suggested."

5/5: "**Great**, I got items I want to watch from the **first** recommended items..."

Interpretation: Even critical feedback was constructive, genuine and continuous engagement.

“User-controlled, privacy-preserving AI

isn't a future state.

It's a design choice we can make today.”

When you design for control rather than extraction, users respond. They engage. They understand. They prefer it.

Thank You

Let's Give Control Back to Users

Thank you to the 22 participants for trust, time and unlimited support!

Questions? manel@cwi.nl